Dhrumil Bhut

Software Engineer | Backend Systems & AI Infrastructure
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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience owning backend systems end to end in production, designing and shipping distributed systems, async pipelines, and AI infrastructure from scratch. Most effective when I own a problem from schema to deployment, not just a feature inside an existing monolith.

WORK EXPERIENCE

Software Engineer 08/2023 to Present
Zuru Tech India Pvt. Ltd., Ahmedabad, India

- Built the brand asset library end to end from scratch: nested folders with move, copy, and cascading delete, multi-filter media retrieval via CTEs that cut query execution time by 40% and improved performance by 20%, watermarking with 6 placement options, all-or-none bulk upload, subtitle support, SKU filtering, and role-based access.
- Designed the schema and dual state machine for a multi-party licensor approval workflow from scratch: separate status lifecycles for projects and individual assets, hybrid core columns and JSONB for dynamic per-category fields, bulk asset review actions, and role-based write access enforced at the brand level.
- Built the async media publishing pipeline end to end from scratch using BullMQ and Redis for distributed job scheduling with deduplication, retry logic, and concurrency control, and RabbitMQ consumer pipelines with cron schedulers to handle media publishing across Instagram, TikTok, and YouTube entirely outside the request lifecycle.
- Built a dedicated webhook ingestion service from scratch that receives real-time events from Instagram and TikTok, publishes them into RabbitMQ, and fans out to multiple internal consumers independently without coupling to the external API contract.
- Architected serverless media processing on AWS: preview generation via Lambda cut compute costs by ~20% and load times by ~30%. Built an S3 photo caching layer for creator profiles that copies on save and deletes on remove, solving CDN expiry for third-party marketplace photos.
- Designed dual-layer rate limiting with two independent middleware chains for inbound API traffic and outbound YouTube API quota. Implemented Microsoft Azure AD SSO end to end: access-token validation, email-domain enforcement, and edge case handling for disabled and deleted users.

PROJECTS

BehavTest *(Behavioral Regression Testing for AI Applications)* github | docs | npm

- Open source behavioral regression testing tool for LLM apps, agents, and RAG pipelines. Shipped as a product: published on npm, listed on the GitHub Marketplace, CI green on Ubuntu, macOS, and Windows, versioned releases and a full documentation site.
- Core engine runs each test case through the real pipeline multiple times and compares against a baseline using statistically honest methods: Wilson confidence intervals, Fisher exact test, and case-stratified permutation tests. Reports not significant when the data does not support a conclusion rather than forcing a verdict.
- Ships with RAG scorers (faithfulness at claim and answer level, retrieval with hit/recall/precision/MRR, contextRelevance per chunk), a local dashboard with run comparison and calibration views, a GitHub Action for CI, and a fire-and-forget SDK that adds zero latency to the host pipeline.

ContextLens *(Self-Hosted RAG Observability Tool)* github | live

- Built end to end: Python SDK, FastAPI ingest API, Celery worker for async claim processing, PostgreSQL with pgvector, Redis, and a Next.js dashboard. Full Docker Compose deployment with a landing page and documentation site.
- Pipeline decomposes every LLM response into atomic claims, embeds claims and source chunks in a single batched API call, attributes each via cosine similarity, and classifies failures as retrieval failures (chunk never fetched) or generation failures (LLM misrepresented a chunk it had). Validated on a real 14-query RAG run: initial 0.75 threshold misclassified paraphrased claims; fixed with a three-band confidence model and separate refusal detection.

PayFlow *(Payment Processing System)* github

- Async payment processing system with Node.js, RabbitMQ, PostgreSQL, and Redis. Two-layer locking via Redis SET NX and PostgreSQL FOR UPDATE ensures no payment is double-processed under concurrent worker load.
- Circuit breaker for fast-fail on gateway outages, exponential backoff retries via Dead Letter Exchange, and a crash-recovery watchdog for zero stuck payments under failure. Deployed with 47 unit and integration tests covering state transitions, race conditions, and webhook conflict resolution.

SKILLS

Languages	Python | JavaScript | TypeScript | SQL
Backend & API	Node.js | Express.js | FastAPI | AWS Lambda | Azure AD SSO | S3
Queues & Jobs	RabbitMQ | BullMQ | Redis | Dead Letter Exchange | Celery
AI & ML Infrastructure	RAG Pipelines | pgvector | LLM Evaluation | OpenAI API | Statistical Testing
Databases	PostgreSQL | pgvector | SQLite | Redis | Query Optimization | Schema Design
Infrastructure & DevOps	Docker | Docker Compose | GitHub Actions | AWS | CI/CD
Testing	Jest | Unit Testing | Integration Testing | Regression Testing

EDUCATION

B.E. Computer Science & Engineering 2020 to 2024
Darshan Institute of Engineering and Technology, Rajkot, India CGPA: 8.74

CERTIFICATES

IBM AI Engineering, IBM (Coursera) Machine Learning Specialization, deeplearning.ai (Coursera)